

Safe Operating Procedures

Stationary Engines- Petrol/Diesel

Various locations

TOI-OHOMAI
Institute of Technology

Te Pūkenga

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PERSONAL PROTECTIVE EQUIPMENT



Version: 1	Key Contact: Eli Crawshaw
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HAZARDS ASSOCIATED WITH THIS EQUIPMENT

PERMANENT HAZARDS	List	Control Method	Lock out Yes or No
Electricity	Electric shock, electrical burn	Remind students and tutors to be situationally aware when working on the car electrical circuit, where necessary ensure the car is turned off and the battery is isolated. Teach fundamentals of how electricity works on running engines.	
Moving parts	Physical injuries including tears, crushing and loss of limbs	Remind students and tutors to be situationally aware when working around vehicles and in the engine bay. Explain the importance of high-pressure lines and how to correctly work on them. Demonstrate and use correct starting procedures.	
Manual handling	Material Safety Data	Place vehicle at correct heights to remove components and use transmission jack to assist in removing and installing of components as required. Ensure hoist is in ergonomically correct position to work on.	
Sharp objects	Physical injuries, cuts and splinters	Remind students and tutors to be situationally aware.	
Slips and trips	Physical injuries	Ensure good housekeeping. Sweep, clean and tidy as you go, don't leave rubbish on ground, mop up spills etc.	
Noise	Hearing damage	Use in a well-ventilated area and wear appropriate PPE.	
Hazardous substances	Skin, eye, and respiratory irritation	Use in a well-ventilated area and wear appropriate PPE.	
Dust	Skin, eye, and respiratory irritation	Use in a well-ventilated area when using compressed air, blow outside where possible. Use appropriate PPE including masks.	
Medical conditions	Any person with a medical condition including injury, epilepsy etc	Remind students and tutors NOT to operate a motor vehicle while on prescription medicine that could legally impair their alertness or judgment. Students are to inform tutors if they are on prescription medication.	
Hazardous materials	Skin, eye, and respiratory irritation, chemical burns	Use in a well-ventilated area and wear appropriate PPE.	
Fatigue	Crash, physical injury	Remind students and tutors NOT to operate a motor vehicle if run down or tired as it affects alertness or judgment.	
Flammable	Explosions or Fire	Use in a well-ventilated area and wear appropriate PPE. Ensure correct draining procedure for all flammable liquids. Have fire extinguishers present. Indicate where they are located.	

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PURPOSE/PROCESS



PROCEDURES:

DO NOT USE THIS MACHINE UNLESS A TUTOR HAS INSTRUCTED AND TRAINED YOU IN ITS SAFE USE AND OPERATION

Pre-Start Check

1. DO NOT use this machine unless you have proven to your trainer that you are competent in its safe use and operation.
2. Locate and ensure you are familiar with all engine operations and controls.
3. Tutors are to ensure previous work is double checked and is in working order before students start working on the training vehicles.
4. Ensure the equipment you will be using to work on the engine has been properly maintained.
5. Ensure the area is clean and clear of grease and oil.
6. Check workspaces and walkways to ensure no slip/trip hazards are present.
7. Check all safety devices are in good condition.
8. Ensure the work area you will be working in is well-ventilated. Fit extraction system and start the fume extraction unit before running the vehicle.
9. Ensure all flammable materials are correctly stored before operating.
10. Do NOT operate an engine after consuming alcohol or other substances (including prescription and non-prescription medications) that could legally impair their alertness or judgment.
11. Do NOT use a handheld cell phone or other portable electronic device while operating the engine.
12. Perform a pre-operational check of the engine.
13. Students are to stand clear when engine is been started.
14. Ensure fuel Line is connected to fuel tank.

Operation

1. Only one person shall operate the engine at a time. Ensure the area is clear of people and equipment before operating or working on it.
2. During operation and when cooling down, be aware that parts of the engine are hot.
3. When working on the engine and if the engine is running, stay clear of any moving parts including drivebelts or any moving shafts.
4. During operation, be aware when working on the cooling system, that it is under pressure and the coolant and water is hot, ensure that you let it cool down before removing the radiator cap.
5. During operation and when working on the high-pressure lines of the diesel engine, depressurize the system first.
6. When charging the diesel engine, ensure the battery is ventilated and isolated from the engine.
7. Switch off the engine when work is completed.
8. Keep the equipment and work area safe, clean, and tidy.
9. Ensure two-way communication with the tutor or another student before starting the engine.

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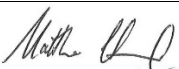
- Person to shout out starting engine
 - The other person to shout out all clear
10. Ensure the engine is clear before starting the vehicle.
 11. Ensure appropriate PPE is used when working on the engine.
 12. When working on vehicle engine and it is not operating, ensure key is removed from ignition and is put on the dash or seat, and battery is disconnected.

Close Down

1. Switch off the engine when work is completed.
2. Ensure battery isolation switch is switched off or battery is disconnected and put away.

Routine maintenance

Lock out and/or isolate any equipment before inspection or maintenance.

	Name	Title	Signature	Date
Author	Elli Crawshaw	Area expert		07/02/2024
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Document Control

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